Is fermion parity protected by topology under noise, and how does the chain length L matter?

Seminar notes (Block 4 – the professor's questions)

July 6, 2026

Abstract

The professor asked whether the fermion parity of the Kitaev-chain simulation is "protected by topology," and how the chain length L enters. We answer both in the noisy setting built in Week 9, applying Qiskit's single-qubit decoherence channels to the topological ground state. The short answers: (Q1) what is conserved is conserved by *symmetry*, not topology — phase damping keeps $\langle \hat{P} \rangle$ exactly fixed because it is parity-preserving, while the genuinely topological object, the non-local edge string, is not noise-immune at any fixed L; (Q2) the chain length acts in *opposite* directions — the intrinsic parity gap is protected exponentially, $\Delta_P(L) \sim e^{-L/\xi}$, but the vulnerability to T_1 poisoning and to depolarizing contrast loss *grows* with L.

1 Background: symmetry versus topology

Under Jordan–Wigner the Kitaev chain is

$$H = -\frac{\mu}{2} \sum_j Z_j + \frac{t-\Delta}{2} \sum_j X_j X_{j+1} + \frac{t+\Delta}{2} \sum_j Y_j Y_{j+1}, \qquad \hat{P} = \prod_j Z_j. \tag{1}$$

Every term has an even number of X/Y factors, so $[H, \hat{P}] = 0$ in *every* phase: parity conservation is a $\mathbb{Z}_2$ symmetry of the Hamiltonian, not a topological effect. What topology adds is that in the topological phase ($|\mu| < 2t$) the two Majorana ends fuse into one non-local fermion $f = \frac{1}{2}(\gamma_L + i\gamma_R)$ whose occupation is the parity bit; the even/odd ground states become degenerate with splitting (the *parity gap*)

$$\Delta_P(L) = \left| E_0^+ - E_0^- \right| \;\approx\; 2t(1-\lambda^2)\lambda^L, \qquad \lambda = \frac{|\mu|}{2t} = e^{-1/\xi}. \tag{2}$$

Because f joins the two ends, no *local* operator can read or flip the bit. The question for Block 4 is what survives once the device decoheres.

2 Which noise channels respect parity

A single-qubit channel matters for $\hat{P}$ only through how its Kraus operators commute with the local Z_j. Using Qiskit's standard channels:

Channel (Qiskit)	Acts through	vs. $\hat{P} = \prod_j Z_j$	Effect on $\langle \hat{P} \rangle$
`phase_damping_error` (T_2)	Z_j	commutes (preserving)	exactly conserved
`depolarizing_error`	X_j, Y_j, Z_j	X, Y anticommute	shrinks $(1-p)^L$ toward 0
`amplitude_damping_error` (T_1)	σ_j^-	violating, biased to $\|0\rangle$	leaks to odd sector

Z_j commutes with $\hat{P}$, so dephasing leaves $\langle \hat{P} \rangle$ untouched. X_j, Y_j and the lowering operator $\sigma^- = \frac{X+iY}{2}$ anticommute with Z_j, so a single such event flips the global parity. Amplitude damping is additionally *biased*: it preferentially sends $|1\rangle \to |0\rangle$. In the repository Jordan–Wigner convention $c_j^\dagger c_j = (I + Z_j)/2$, so $|0\rangle$ is the occupied local state; the essential point is that the channel changes the local Z_j eigenvalue and can therefore leak the protected even state into the odd sector rather than scrambling it symmetrically.

Connection to Week 9. The realistic `thermal_relaxation_error` (T_1, T_2, τ) used in the Week 9 gate-noise study is exactly the composition of these two pieces: its T_1 part is the parity-*violating* amplitude damping, its T_2 part is the parity-*preserving* dephasing. So the two questions below dissect the same physical channel that drives the circuit-level model.

3 Q1: is the parity protected by topology, under noise?

We prepare the exact even-parity ground state at $\mu = t$ (topological), apply each channel at increasing strength p, and read off $\langle \hat{P} \rangle$ and the edge string (Fig. 1). Two different objects behave very differently.

The conserved quantity $\langle \hat{P} \rangle$ — protected by *symmetry*. Under phase damping $\langle \hat{P} \rangle$ stays *exactly* 1 (flat to machine precision), because the Kraus operators commute with $\hat{P}$. But this is the $\mathbb{Z}_2$ symmetry, which holds in every phase — it is *not* topology. Both parity-violating channels break it: depolarizing drives $\langle \hat{P} \rangle \to 0$ as $(1-p)^L$, and amplitude damping leaks into the odd sector ($\langle \hat{P} \rangle$ falling, but staying above depolarizing because the bias pulls back toward the $\hat{P} = +1$ vacuum).

The topological order parameter — *not* noise-immune. The non-local edge string decays under *every* channel, *including* the parity-preserving phase damping that leaves $\langle \hat{P} \rangle = 1$: dephasing destroys the cross-chain coherence the string relies on, so its contrast falls even while parity is perfectly conserved.

Answer. Parity conservation is protected against parity-preserving noise — but by *symmetry*, not topology — and is destroyed by parity-violating noise (T_1 poisoning, depolarizing). The genuinely *topological* object, the non-local edge string, is *not* protected against any noise at fixed L: conserving parity does not preserve the topological signal. Topology does not buy fixed-L noise immunity; where it enters is the chain-length scaling.

4 Q2: how does the chain length change this, under noise?

Length acts in two competing ways (Fig. 2).

- **Intrinsic protection improves with L.** Noiselessly, $\Delta_P(L) \sim e^{-L/\xi}$ in the topological phase ($\mu = t$): on a semilog axis a straight line of slope $\ln(|\mu|/2t)$, falling from 0.41 at $L = 2$ to 5.9×10^{-3} at $L = 8$. In the trivial phase ($\mu = 3t$) the gap is flat at $\mathcal{O}(1)$ — there is no degeneracy to protect.
- **Vulnerability worsens with L.** At fixed channel strength $\gamma = p = 0.1$, the odd-sector leakage under amplitude damping grows from 0.03 to 0.21 (more sites that can lose a fermion, $\sim 1 - (1-\gamma)^L$), and the edge-string loss under depolarizing grows from 0.19 to 0.57 (a longer non-local string accumulates more single-qubit contrast loss).

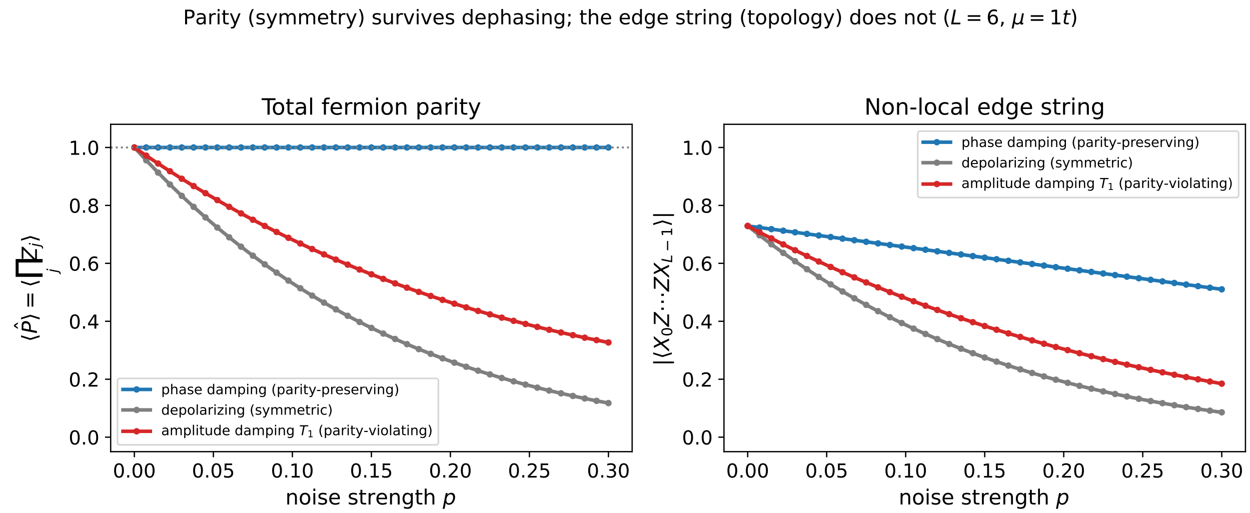

Figure 1: Topological state at $L = 6$, $\mu = t$, Qiskit single-qubit channels. Left: $\langle \hat{P} \rangle$ stays exactly 1 under phase damping (symmetry), collapses under depolarizing, and leaks (stays biased high) under amplitude damping. Right: the non-local edge string decays under *all* channels — including the parity-preserving phase damping — so parity conservation does not protect the topological signal.

Answer. There is no monotone "longer is better." The topological degeneracy is exponentially better protected at large L, but the noisy device loses parity and contrast faster at large L. The two trends oppose, defining an optimal working length — the direct analogue, in the chain-length axis, of the optimal ansatz depth found in Week 9.

5 Summary

- Parity *conservation* is a $\mathbb{Z}_2$ symmetry, not topology; it survives parity-preserving (T_2) noise and is broken by parity-violating (T_1, depolarizing) noise.
- The *topological* object (the non-local edge string / the degeneracy) is not noise-immune at fixed L — every channel erodes it.
- Topology's role is in the L-scaling: intrinsic protection improves as $e^{-L/\xi}$ while the noisy device degrades with L, giving an optimal length — the chain-length counterpart of the Week 9 optimal-depth result.
- These three axes — noise strength, ansatz depth (Week 9), and chain length — are the knobs that decide whether a NISQ measurement of the Kitaev chain can still resolve the topological phase.

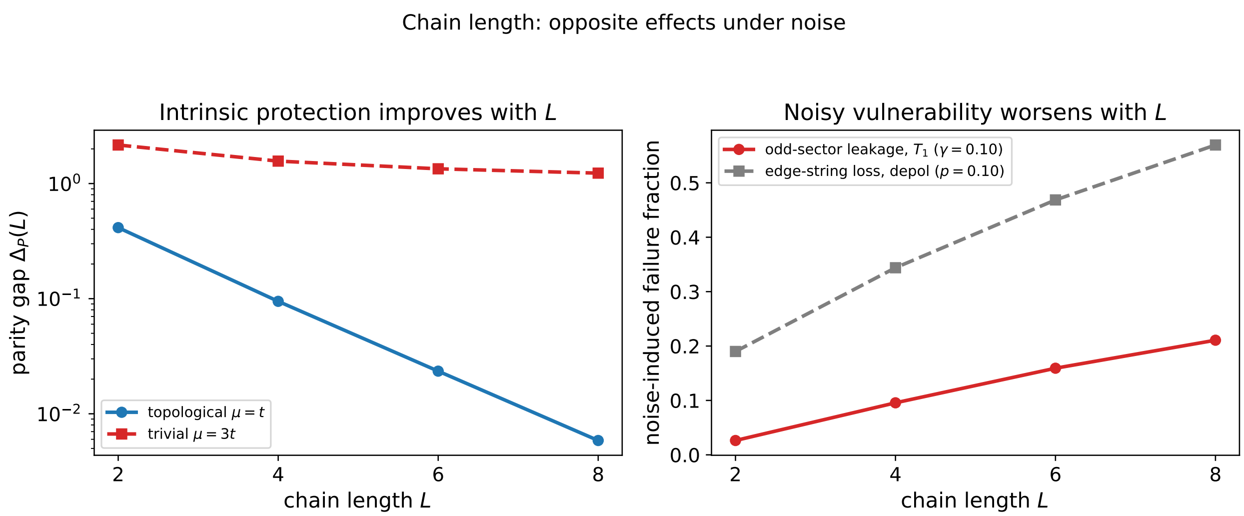

Figure 2: Left: noiseless parity gap $\Delta_P(L)$ — exponential decay in the topological phase (intrinsic protection improving with L) versus the flat $\mathcal{O}(1)$ trivial gap. Right: at fixed noise $\gamma = p = 0.1$, odd-sector leakage under T_1 and edge-string loss under depolarizing both grow with L. Longer chains are intrinsically better protected but noisier to operate.